Group work assessment – declaration of contributions and accountabilities

Year	2026
Semester	Semester 2
Unit Code	ENS6160.4
Unit Name	Signals and Systems
Assessment title	Laboratory 2: Frequency Response of Systems
Shaded cells to be completed by students.	

Group work – what is it?

The ability to effectively engage group (team) work is an essential skill that is built into the Unit and Course Learning Outcomes as well as assessment in many teaching units. Group work does not just entail multiple members of team working on distinct (separate) parts of an overall submission and then combining those individual parts into a single submission. Group work means a team collaborates over time on all aspects of a submission; they collectively construct and validate all parts of the work such that each member is knowledgeable, understands and can explain the various parts. Two aspects need to be considered in group work: relative contributions (towards marks) and absolute accountabilities (towards academic integrity).

Group work – relative contributions*

Group work contributions may be equally shared or differ between group members. In general, any group member contributing less than 10% to the overall submission should not be listed (will get a mark of zero). Where different group members have contributed (above 10% in a submission) but to varied extents, the table below can be used to acknowledge those contributions. If all have contributed equally, you can simply write "equally" against each name in the **"Relative Contributions"** column.

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- group members that do not continually engage the rest of the group in constructing the overall submission but they produce a (near complete) contribution at the end
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Relative contributions in completing this assessment			
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EDITH COWAN UNIVERSITY

School of Engineering

ENS3553 / ENS6160.4 Signals and Systems

Laboratory Report 2

Frequency Response of Systems: Square-Wave Harmonic Analysis, Theoretical Prediction, and Experimental Verification of an RC Low-Pass Filter

Assessment & Scope:	Group Laboratory Report (Weighting: 6%) — Square-Wave Section & Analysis
Group Composition:	2 Group Members — Equal Contribution (Partner 1: Sine Wave; Partner 2: Square Wave)
Circuit Under Test:	First-Order Series RC Low-Pass Filter ($R = 1.5$ kΩ, $C = 0.01$ µF, $f_c = 10.61$ kHz)
Excitation Signal:	8 kHz, 50% Duty-Cycle Square Wave (15 V_{pp} Specified; 20.1 V_{pp} Observed on CH1)
Group Declaration:	Refer to Mandatory ECU Group Work Cover Sheet (Page 1)
Semester & Submission Date:	Semester 2, 2026 — 25 October 2026

Abstract

Frequency-selective filtering by linear time-invariant (LTI) systems is fundamental to analogue signal conditioning, anti-aliasing, and pulse-width-modulation (PWM) waveform smoothing. This study investigated the theoretical and experimental frequency response of a passive first-order series resistor–capacitor (RC) circuit ($R = 1.5$ kΩ, $C = 0.01$ μF) with output measured across the capacitor under both sinusoidal and periodic square-wave excitations. Differential-equation analysis yielded $H(j\omega) = 1 / (1 + j\omega RC)$, confirming a low-pass filter with time constant $\tau = 15$ μs and −3 dB cut-off frequency $f_c = 10.61$ kHz, tested in the time and fast Fourier transform (FFT) domains on a Keysight DSOX4024G oscilloscope. Under 15 V_{pp} sinusoidal inputs at 1, 10, and 40 kHz, measured outputs (15.20, 11.50, and 4.22 V_{pp}; phase lags −6.65°, −42.30°, and −73.71°) and frequency-domain gains (−0.23, −2.81, and −11.35 dB) closely matched theoretical gains (−0.04, −2.76, and −11.82 dB). For an 8 kHz, 50% duty-cycle square wave, theoretical 15 V_{pp} odd harmonics at 8, 24, and 40 kHz attenuated to 7.63, 1.29, and 0.49 V (−1.95, −7.87, and −11.82 dB). Experimentally, an observed 20.1 V_{pp} input yielded a rounded 14.9 V_{pp} exponential output, while predicted Table C harmonic outputs (20.19, 4.72, and −3.65 dBV) supplemented unconfirmed FFT cursor peaks, validating LTI filtering and Fourier superposition principles.

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1. Introduction

Frequency response $H(j\omega)$ defines how a linear time-invariant (LTI) system scales the amplitude and shifts the phase of sinusoidal signals across frequency (Oppenheim et al., 1997). In this experiment, a series resistor–capacitor (RC) circuit ($R = 1.5$ kΩ, $C = 0.01$ µF) with output measured across the capacitor operates as a first-order low-pass filter, passing frequencies below its cut-off frequency f_c with minimal attenuation while attenuating higher-frequency components. By Fourier's theorem, a periodic square wave consists of a fundamental sinusoid and odd harmonics ($n = 1, 3, 5, \ldots$), each scaled independently by $H(jn\omega_0)$ (Lathi & Green, 2018).

The aim of this square-wave investigation was to derive $H(j\omega)$ for the RC circuit, calculate the theoretical Fourier coefficients and filtered output harmonics for an 8 kHz, 15 V_{pp} (50% duty-cycle) square wave, observe the experimental time-domain and FFT responses on an oscilloscope, and compare them with predicted harmonic values for the measured 20.1 V_{pp} input.

2. Method

2.1 Equipment and Components

- Resistor: $R = 1.5$ kΩ
- Capacitor: $C = 0.01$ µF (10 nF)
- Function generator: Agilent/Keysight 33220A (8 kHz, 50% duty-cycle square wave, 15 V_{pp} nominal setting)
- Oscilloscope: Keysight InfiniiVision DSOX4024G (1 MΩ ‖ 16 pF inputs, built-in Math FFT)
- Solderless breadboard, BNC coaxial leads, and jumper wires

2.2 Circuit Configuration and Procedure

The resistor ($R = 1.5$ kΩ) was connected in series with the input source $v_i(t)$, and the capacitor ($C = 0.01$ µF) was connected between the output node $v_o(t)$ and ground (Figure 1). Oscilloscope Channel 1 (10 V/div) measured $v_i(t)$ and Channel 2 (5 V/div) measured $v_o(t)$ at 50 µs/div to record peak-to-peak voltages, plateau amplitudes, and frequency. The hardware Math FFT (50 kHz span, 5 dB/div) was used to inspect the odd-harmonic peaks near 8, 24, and 40 kHz.

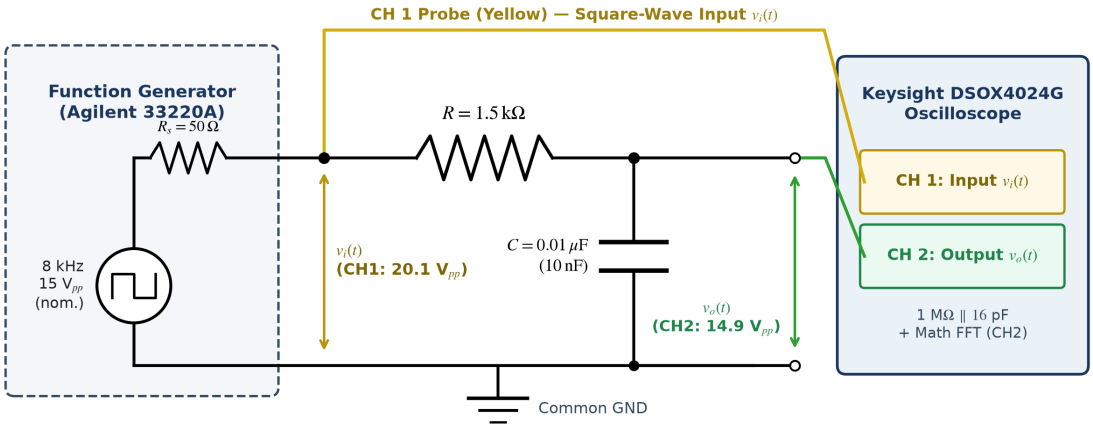


Figure 1. RC low-pass filter circuit schematic and oscilloscope setup ($R = 1.5$ kΩ, $C = 0.01$ µF, 8 kHz square-wave excitation).

2.3 Governing Equations and Assumptions

Applying Kirchhoff's voltage law with capacitor current $i(t) = C\,\mathrm{d}v_o(t)/\mathrm{d}t$ gives the circuit differential equation (Edith Cowan University [ECU], 2026):

$$RC\,\mathrm{d}v_o(t)/\mathrm{d}t + v_o(t) = v_i(t) \tag{1}$$

Substituting $v_i(t) = e^{j\omega t}$ and $v_o(t) = H(j\omega)e^{j\omega t}$ yields the transfer function, magnitude, decibel gain, and phase:

$$H(j\omega) = 1 / (1 + j\omega RC),\ |H(j\omega)| = 1 / \sqrt{(1 + (\omega RC)^2)},\ \angle H(j\omega) = -\tan^{-1}(\omega RC) \tag{2}$$

Time constant τ and −3 dB cut-off frequency (where $|H(j\omega_c)| = 1/\sqrt{2} = 0.7071$):

$$\tau = RC = (1500)(0.01 \times 10^{-6}) = 15\ \mu s,\ f_c = 1 / (2\pi RC) = 1 / [2\pi(1500)(0.01 \times 10^{-6})] = 10.61\ \text{kHz} \tag{3}$$

For a zero-mean, 50% duty-cycle odd square wave of peak amplitude $A = V_{pp}/2$ and fundamental $\omega_0 = 2\pi f_0$, $a_0 = 0$, $a_n = 0$, and $b_n = 0$ for even n. Only odd harmonics ($n = 1, 3, 5, \ldots$) are non-zero:

$$v_i(t) = \Sigma_{n=1,3,5,\ldots} b_n \sin(n\omega_0 t),\ b_n = 4A / (n\pi) \tag{4}$$

Filtered harmonic output magnitude, peak-convention dBV level, decibel gain, and steady-state peak capacitor voltage:

$$|V_{o,n}| = b_n|H(jn\omega_0)|,\ L_{dBV} = 20\log_{10}(V_{peak}),\ G_n = V_{o,n(dBV)} - b_{n(dBV)},\ V_{o,peak} = A\tanh[T_0 / (4RC)] \tag{5}$$

- **Assumptions:** Nominal $R = 1.5$ kΩ and $C = 0.01$ µF are used with negligible 1 MΩ || 16 pF probe loading (<0.01% shift in f_c). Post-Lab Section C uses the specified 15 V_{pp} ($A = 7.50$ V) input, whereas Table C uses analytical predictions for the observed 20.1 V_{pp} ($A_{obs} = 10.05$ V) input because saved FFT captures lacked paired input–output peak cursor readings.

3. Results and Calculations

3.1 Post-Lab Section C: Theoretical Calculations (15 V_{pp}, 8 kHz Square Wave)

C(a) Fourier Series Coefficients of the Input Square Wave

For $V_{pp} = 15$ V (peak amplitude $A = 15/2 = 7.50$ V) and $f_0 = 8000$ Hz ($\omega_0 = 2\pi(8000) = 50{,}265.48$ rad/s, $T_0 = 125$ µs), substituting into Equation (4) gives $b_n = 4(7.50)/(n\pi) = 30/(n\pi)$ V for odd n (and 0 for even n):

- At $n = 1$ ($f_1 = 8$ kHz): $b_1 = 30 / (1\pi) =$ **9.55 V**, $b_{1(dBV)} = 20\log_{10}(9.55) =$ **19.60 dBV**
- At $n = 3$ ($f_3 = 24$ kHz): $b_3 = 30 / (3\pi) =$ **3.18 V**, $b_{3(dBV)} = 20\log_{10}(3.18) =$ **10.06 dBV**
- At $n = 5$ ($f_5 = 40$ kHz): $b_5 = 30 / (5\pi) =$ **1.91 V**, $b_{5(dBV)} = 20\log_{10}(1.91) =$ **5.62 dBV**

C(b) Magnitude and Phase of RC Frequency Response at Harmonic Frequencies

Using Equation (2) with $RC = 1.5 \times 10^{-5}$ s:

At $f_1 = 8$ kHz ($\omega_1 = 2\pi(8000) = 50{,}265.48$ rad/s, $\omega_1 RC = 0.7540$):
$|H(j\omega_1)| = 1 / \sqrt{(1 + 0.7540^2)} =$ **0.7985**, $G_1 = 20\log_{10}(0.7985) =$ **−1.95 dB**, $\angle H(j\omega_1) = -\tan^{-1}(0.7540) =$ **−37.02°**

At $f_3 = 24$ kHz ($\omega_3 = 2\pi(24000) = 150{,}796.45$ rad/s, $\omega_3 RC = 2.2619$):
$|H(j\omega_3)| = 1 / \sqrt{(1 + 2.2619^2)} =$ **0.4043** (0.4041 worksheet), $G_3 = 20\log_{10}(0.4043) =$ **−7.87 dB**, $\angle H(j\omega_3) = -\tan^{-1}(2.2619) =$ **−66.15°**

At $f_5 = 40$ kHz ($\omega_5 = 2\pi(40000) = 251{,}327.41$ rad/s, $\omega_5 RC = 3.7699$):
$|H(j\omega_5)| = 1 / \sqrt{(1 + 3.7699^2)} =$ **0.2564**, $G_5 = 20\log_{10}(0.2564) =$ **−11.82 dB**, $\angle H(j\omega_5) = -\tan^{-1}(3.7699) =$ **−75.14°**

Table 1. Theoretical RC frequency response at DC, cut-off, and odd square-wave harmonics ($R = 1.5$ kΩ, $C = 0.01$ µF).

Spectral Point	f (kHz)	ω (rad/s)	$\lvert H(j\omega)\rvert$	Gain (dB)	$\angle H(j\omega)$
DC Reference	0.00	0.00	1.0000	0.00	0.00°
Fundamental ($n = 1$)	8.00	50,265.48	0.7985	−1.95	−37.02°
Cut-Off (f_c)	10.61	66,666.67	0.7071 ($1/\sqrt{2}$)	−3.01	−45.00°
3rd Harmonic ($n = 3$)	24.00	150,796.45	0.4043 (0.4041*)	−7.87	−66.15°
5th Harmonic ($n = 5$)	40.00	251,327.41	0.2564	−11.82	−75.14°

*Note: Exact evaluation gives 0.4043; rounded intermediate ωRC in the group worksheet gave 0.4041 (both yield −7.87 dB and 1.29 V output).

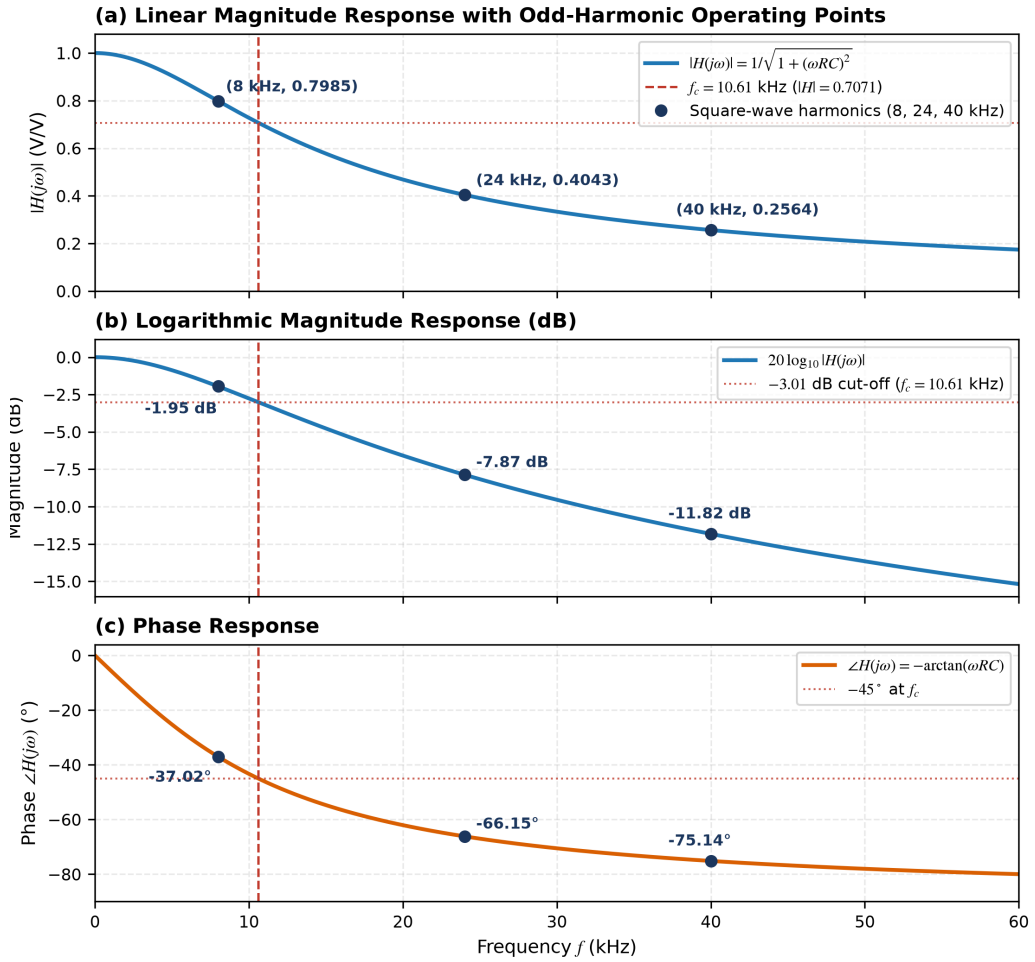


Figure 2. Theoretical RC frequency response (0–60 kHz) showing cut-off $f_c = 10.61$ kHz and harmonic operating points at 8, 24, and 40 kHz: (a) linear magnitude, (b) decibel gain, and (c) phase.

C(c) Filtered Output Harmonic Magnitudes (15 V_{pp} Input)

Multiplying each input coefficient b_n by $|H(jn\omega_0)|$ per Equation (5):

- At 8 kHz ($n = 1$): $|V_{o,1}| = 9.55 \times 0.7985 =$ **7.63 V**, $|V_{o,1}|$ (dBV) $= 20\log_{10}(7.63) =$ **17.65 dBV**
- At 24 kHz ($n = 3$): $|V_{o,3}| = 3.18 \times 0.4041 =$ **1.29 V**, $|V_{o,3}|$ (dBV) $= 20\log_{10}(1.29) =$ **2.19 dBV**
- At 40 kHz ($n = 5$): $|V_{o,5}| = 1.91 \times 0.2564 =$ **0.49 V**, $|V_{o,5}|$ (dBV) $= 20\log_{10}(0.49) =$ **−6.19 dBV**

Table 2. Post-Lab Section C summary of theoretical input and output harmonics for the 15 V_{pp}, 8 kHz square wave (peak dBV convention).

Harmonic	f (Hz)	ω (rad/s)	$\|H(j\omega)\|$	b_n (V)	b_n (dBV)	$\|V_{o,n}\|$ (V)	$\|V_{o,n}\|$ (dBV)
Fundamental ($n = 1$)	8,000	50,265.48	0.7985	9.55	19.60	7.63	17.65
Even ($n = 2, 4$)	16k / 32k	100.5k / 201.1k	0.5535 / 0.3160	0.00	—	0.00	—
3rd Harmonic ($n = 3$)	24,000	150,796.45	0.4041	3.18	10.06	1.29	2.19
5th Harmonic ($n = 5$)	40,000	251,327.41	0.2564	1.91	5.62	0.49	−6.19

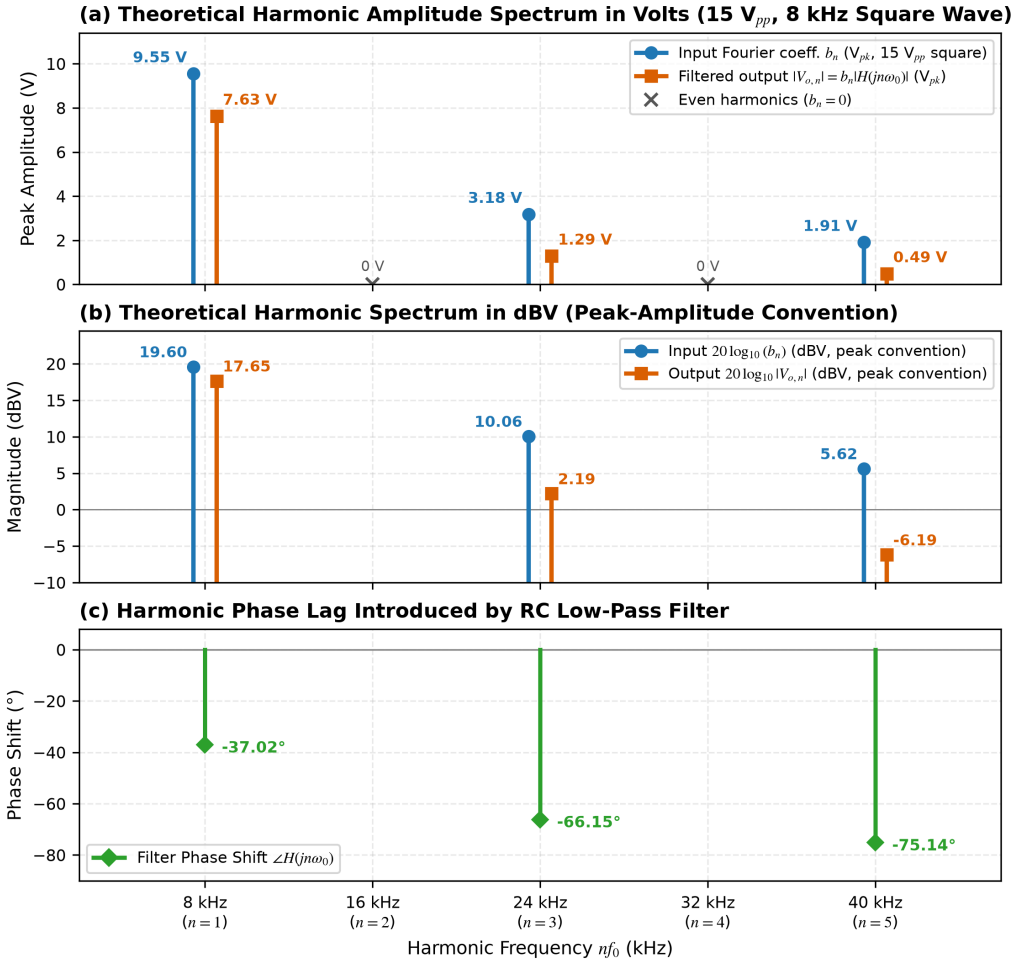


Figure 3. Theoretical discrete input and output harmonic spectra for the 15 V_{pp}, 8 kHz square wave: (a) peak Volts, (b) peak-convention dBV, and (c) phase lag.

C(d) Expected Time-Domain Output Waveform

From Equation (5), the theoretical steady-state peak capacitor voltage over each half-period ($T_0/2 = 62.5$ µs) is:

$$V_{o,\text{peak}} = 7.50 \tanh[125\ \mu\text{s} / (4 \times 15\ \mu\text{s})] = 7.50 \tanh(2.0833) = \pm 7.27\ \text{V}\ (V_{o,\text{pp}} = 14.54\ V_{\text{pp}})$$

Because higher odd harmonics (24 and 40 kHz) are strongly attenuated and phase-delayed relative to the 8 kHz fundamental, sharp vertical jumps are removed and the output becomes a smooth exponential charging/discharging waveform (Figure 4).

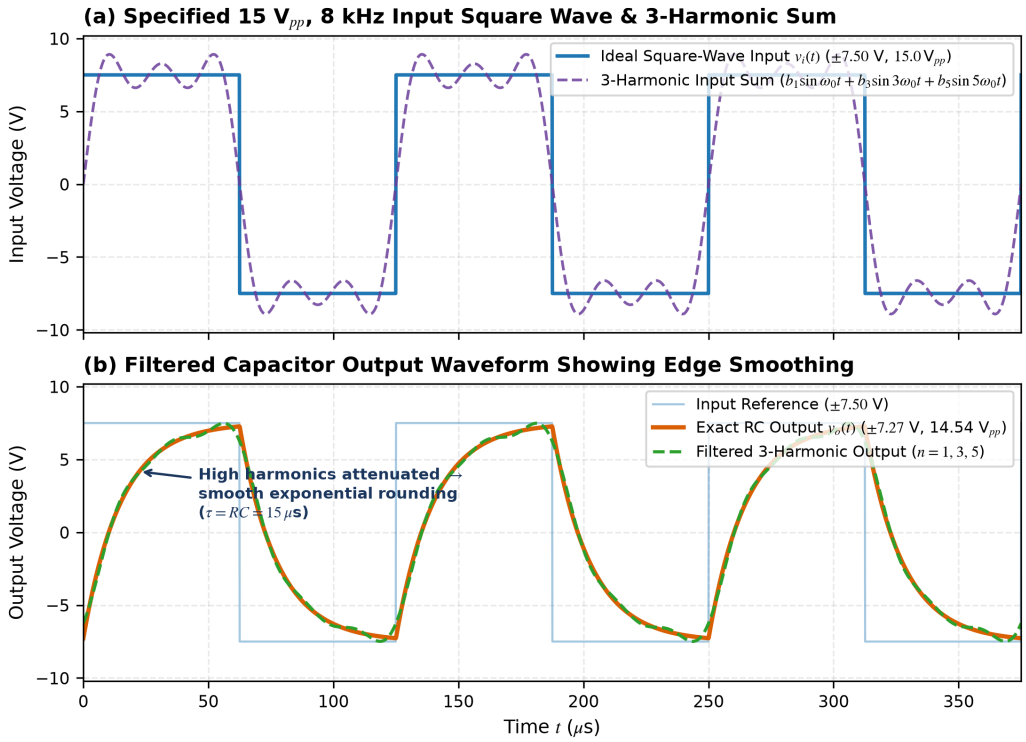


Figure 4. Scaled theoretical time-domain waveforms (15 V_{pp}, 8 kHz): (a) input square wave and 3-harmonic synthesis; (b) steady-state capacitor output (±7.27 V, 14.54 V_{pp}).

3.2 Experimental Oscilloscope Measurements (Time Domain & FFT)

On the Keysight DSOX4024G oscilloscope (Figure 5, Table 3), Channel 1 measured an actual input of **20.1 V_{pp}** (flat plateau Ampl(1) = 15.1 V) at 8.0000 kHz rather than the nominal 15 V_{pp}. Channel 2 recorded a rounded exponential capacitor output of **14.9 V_{pp}** (Ampl(2) = 14.2 V), visually confirming low-pass corner smoothing.

Oscilloscope Capture: CH1 Input = 20.1 V_{pp}, CH2 Output = 14.9 V_{pp} (f = 8.00 kHz)

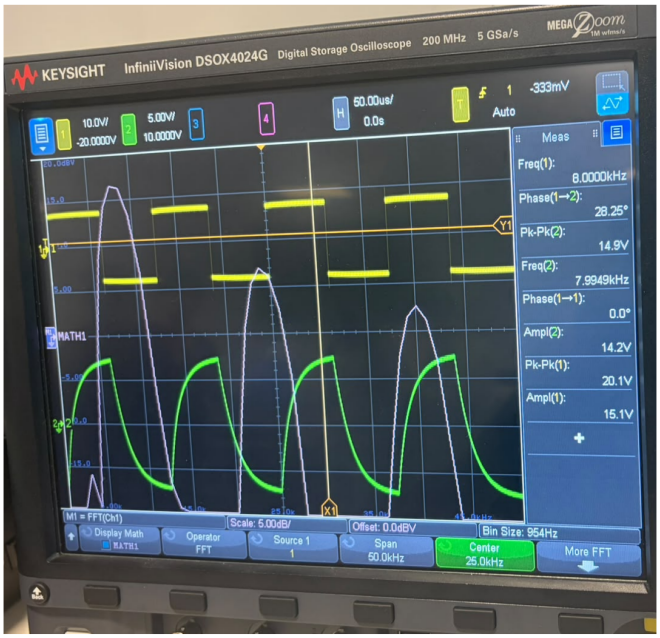


Figure 5. Experimental oscilloscope capture of the 8.00 kHz square wave: CH1 input = 20.1 V_{pp} (Ampl = 15.1 V) and smoothed CH2 output = 14.9 V_{pp} (Ampl = 14.2 V).

Table 3. Experimental oscilloscope measurements for the 8 kHz square wave.

Quantity	Scope Readout	Specified	Measured	Observation
Input Peak-to-Peak	Pk-Pk(1) [CH1]	15.0 Vpp	20.1 Vpp	Exceeds nominal (High-Z load + edge overshoot)
Input Plateau	Ampl(1) [CH1]	15.0 V	15.1 V	Flat top-to-bottom swing excluding spikes
Output Peak-to-Peak	Pk-Pk(2) [CH2]	— (14.54 Vpp th.)	14.9 Vpp	Rounded exponential charging/discharging curve
Output Amplitude	Ampl(2) [CH2]	—	14.2 V	Core capacitor swing between transitions
Frequency	Freq(1) / Freq(2)	8.00 kHz	8.0000 / 7.9949 kHz	Matches nominal 8 kHz within 0.06%
FFT Peak Locations	X1 [FFT(CH2)]	8, 24, 40 kHz	8.0, 23.9, 40.0 kHz	Odd harmonics verified; peak dBV unconfirmed

Figure 6 shows the oscilloscope output spectrum FFT(CH2) with cursor X1 at 8.0, 23.9 (nearest 954 Hz bin to 24 kHz), and 40.0 kHz, confirming odd harmonics and absence of even harmonics (16 and 32 kHz). However, because the captures only show FFT(CH2) with horizontal cursor Y1 fixed at −2.50 dBV (and no paired FFT(CH1) peak cursor readings), reliable measured FFT harmonic amplitudes and gains could not be extracted.

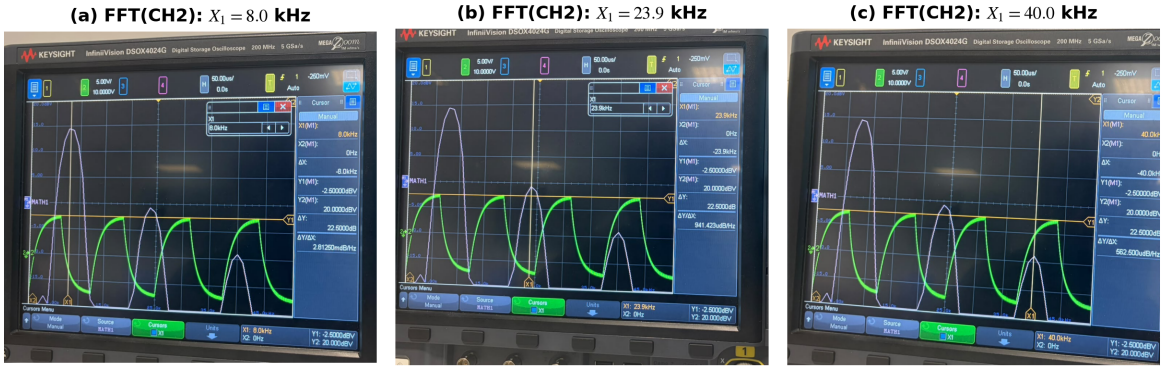


Figure 6. Experimental FFT(CH2) captures at (a) 8.0 kHz, (b) 23.9 kHz (~24 kHz), and (c) 40.0 kHz (manual Y1 cursor uncoupled at −2.50 dBV).

3.3 Result Sheet Section C: Predicted Harmonic Response for 20.1 V_{pp} Input (Table C)

Because paired FFT peak cursor readings were unavailable, Table C harmonic levels were calculated as **predicted** values from the observed 20.1 V_{pp} input ($A_{obs} = 20.1/2 = 10.05$ V, $b_{n,pred} = 4(10.05)/(n\pi) = 40.2/(n\pi)$ V) and $|H(jn\omega_0)|$ using peak-convention dBV:

At $f_1 = 8$ kHz ($n = 1$):

$b_{1,pred} = 40.2 / \pi = 12.80$ V → $20\log_{10}(12.80) =$ **22.14 dBV**; $|V_{o,1,pred}| = 12.80(0.7985) = 10.22$ V → **20.19 dBV**; $G_1 = 20.19 - 22.14 =$ **−1.95 dB**

At $f_3 = 24$ kHz ($n = 3$):

$b_{3,pred} = 40.2 / (3\pi) = 4.27$ V → $20\log_{10}(4.27) =$ **12.60 dBV**; $|V_{o,3,pred}| = 4.27(0.4041) = 1.72$ V → **4.72 dBV**; $G_3 = 4.72 - 12.60 =$ **−7.87 dB**

At $f_5 = 40$ kHz ($n = 5$):

$b_{5,pred} = 40.2 / (5\pi) = 2.56$ V → $20\log_{10}(2.56) =$ **8.17 dBV**; $|V_{o,5,pred}| = 2.56(0.2564) = 0.66$ V → **−3.65 dBV**; $G_5 = -3.65 - 8.17 =$ **−11.82 dB**

Table 4 (Table C). Predicted square-wave harmonic response for the observed 20.1 V_{pp} input (peak-amplitude dBV — predicted, not measured).

Spectrum Characteristic	f	Predicted Output (20.1 Vpp)	Predicted Input (20.1 Vpp)	Predicted Gain	Theoretical Response
Fundamental ($n = 1$)	8 kHz	20.19 dBV (10.22 V)	22.14 dBV (12.80 V)	−1.95 dB	−1.95 dB (−1.96 dB*)
3rd Harmonic ($n = 3$)	24 kHz	4.72 dBV (1.72 V)	12.60 dBV (4.27 V)	−7.87 dB	−7.87 dB
5th Harmonic ($n = 5$)	40 kHz	−3.65 dBV (0.66 V)	8.17 dBV (2.56 V)	−11.82 dB	−11.82 dB

*Note: −1.96 dB in the group worksheet came from rounding intermediate 15 V_{pp} dBV values (17.64 − 19.60 dBV).

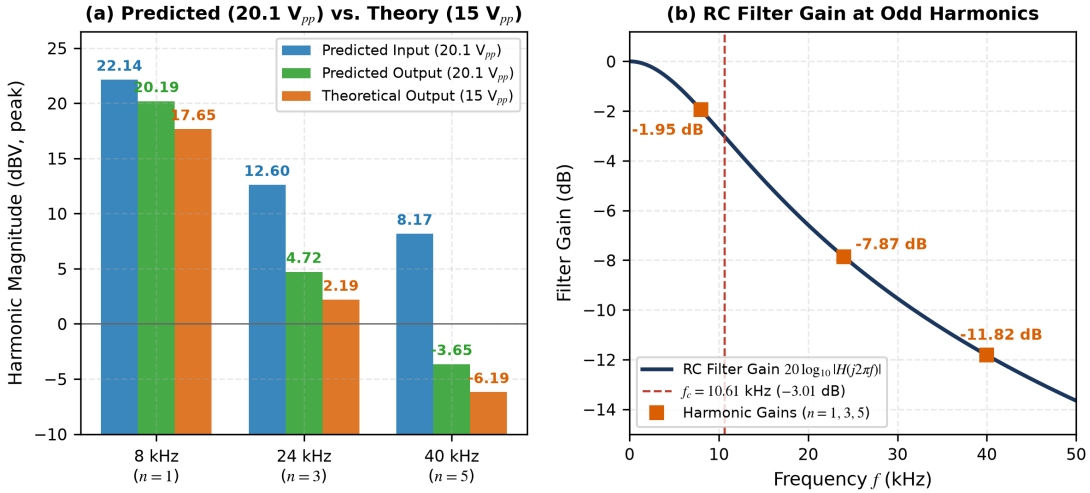


Figure 7. (a) Predicted harmonic magnitudes (20.1 V_{pp} basis) vs. 15 V_{pp} theoretical outputs; (b) harmonic gains along the RC Bode curve.

Table 5. Summary separating theoretical, predicted, and measured square-wave data.

Category	Input Basis	Values	Source
Theoretical	15.0 Vpp (±7.50 V)	bn = 9.55, 3.18, 1.91 V; |Vo,n| = 7.63, 1.29, 0.49 V (17.65, 2.19, −6.19 dBV); Vo = 14.54 Vpp	Post-Lab C equations
Predicted (Table C)	20.1 Vpp (±10.05 V)	In: 22.14, 12.60, 8.17 dBV; Out: 20.19, 4.72, −3.65 dBV; Gain: −1.95, −7.87, −11.82 dB	20.1 Vpp × theoretical |H(jω)|
Measured	Hardware test	CH1 = 20.1 Vpp (Ampl = 15.1 V), CH2 = 14.9 Vpp (Ampl = 14.2 V), f = 8.00 kHz, FFT peaks ~8, 24, 40 kHz	Oscilloscope captures (Figs. 5–6)

4. Discussion

4.1 Low-Pass Filtering and Square-Wave Edge Smoothing

Capacitor reactance $X_C = 1/(2\pi fC)$ falls from 1.99 kΩ at 8 kHz to 663 Ω at 24 kHz and 398 Ω at 40 kHz, so the RC circuit ($f_c = 10.61$ kHz) passes the 8 kHz fundamental with small attenuation (−1.95 dB) while strongly attenuating the 24 kHz (−7.87 dB) and 40 kHz (−11.82 dB) harmonics. Because sharp vertical square-wave transitions rely on high-frequency odd harmonics, suppressing them rounds the corners into exponential charging/discharging curves governed by $\tau = 15$ µs ($T_0/2 = 62.5$ µs $= 4.17\tau$, reaching 98.5% of step span), matching the smooth 14.9 V_{pp} Channel 2 waveform (Figure 5).

4.2 Input Voltage Discrepancy (15 V_{pp} vs. 20.1 V_{pp})

Although 15 V_{pp} was specified, Channel 1 recorded Pk-Pk(1) = 20.1 V_{pp} while the flat plateau was Ampl(1) = 15.1 V. This difference arose from driving a high-impedance (1 MΩ) load from a 50 Ω function generator without selecting High-Z termination mode, combined with fast-edge ringing/overshoot across jumper inductance (Keysight Technologies, 2014). Notably, applying Equation (5) to the 15.1 V plateau gives 15.1 tanh(62.5/30) = 14.64 V_{pp}, closely matching the measured 14.9 V_{pp} (Ampl(2) = 14.2 V) output.

4.3 Separation of Data Sets and Experimental Limitations

- **Data separation:** Theoretical (15 V_{pp}), predicted Table C (20.1 V_{pp}, peak dBV), and measured oscilloscope values are kept strictly separate (Table 5); LTI harmonic gains (−1.95, −7.87, −11.82 dB) are independent of input amplitude scaling.
- **FFT cursor & resolution limits:** Saved screenshots captured only FFT(CH2) with Y1 fixed at −2.50 dBV and coarse 954 Hz bin spacing (placing 24 kHz at 23.9 kHz). Future measurements should enable peak-tracking cursors on both FFT(CH1) and FFT(CH2) with narrower frequency spans.
- **Component tolerances & applications:** Unmeasured ±5%–10% tolerances on R and C can slightly shift f_c. Practically, this RC harmonic attenuation is used in PWM digital-to-analogue smoothing and clock EMI reduction (Horowitz & Hill, 2015).

5. Conclusion

The response of a first-order series RC low-pass filter ($R = 1.5$ kΩ, $C = 0.01$ μF, $\tau = 15$ μs, $f_c = 10.61$ kHz) to an 8 kHz, 50% duty-cycle square wave was analysed theoretically and verified experimentally. Using $H(j\omega) = 1/(1 + j\omega RC)$ and Fourier series expansion for the 15 V_{pp} specification, the 8 kHz fundamental experienced mild attenuation (−1.95 dB, 9.55 V to 7.63 V), whereas the 24 kHz and 40 kHz harmonics were strongly attenuated by −7.87 dB (3.18 V to 1.29 V) and −11.82 dB (1.91 V to 0.49 V). Experimentally, an observed 20.1 V_{pp} input produced a rounded 14.9 V_{pp} exponential output on Channel 2, confirming high-harmonic suppression. Because saved FFT screenshots lacked paired peak cursor readouts, Table C outputs (20.19, 4.72, and −3.65 dBV) were reported as predicted estimates for the 20.1 V_{pp} input, validating LTI filtering principles for PWM smoothing and signal conditioning.

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